WORKED SOLUTION

STEP II 2008, Question 4

Pure • Implicit Differentiation

> **STEP II 2008, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s2-q4

The key idea

The angle between two lines is easiest to handle with vectors: the normal to $F(x,y)=1$ points along ∇F, and for two vectors $\mathbf{p}$ and $\mathbf{n}$ in the plane, tan of the angle between them is |cross product| divided by the dot product. Here the dot product turns out to be exactly the left-hand side of the curve's equation, which is 1. After that, θ is stationary exactly when y^2-x^2 is, and the rest is algebra with x^2+y^2 and xy.

Solution

The gradient

Differentiating $x^2+y^2+2axy=1$ implicitly with respect to x,

$$2x+2y\frac{\mathrm{d}y}{\mathrm{d}x}+2a\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)=0 \quad\Longrightarrow\quad (ax+y)\frac{\mathrm{d}y}{\mathrm{d}x}=-(x+ay).$$

So, provided $ax+y\neq 0$, the gradient is $-\dfrac{x+ay}{ax+y}$, as required.

The angle θ

The tangent has direction $(ax+y,\ -(x+ay))$, so the normal has direction

$$\mathbf{n}=(x+ay,\ ax+y),$$

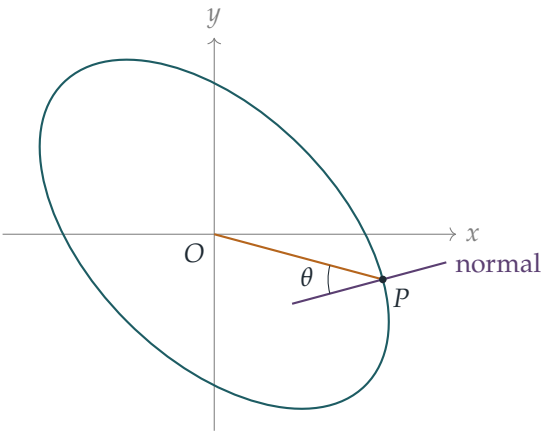
the vector perpendicular to it. (This is $\frac{1}{2}\nabla F$ for $F=x^2+y^2+2axy$, and it makes sense even where $ax+y=0$.) The vector $\overrightarrow{OP}$ is $\mathbf{p}=(x,y)$. Let ϕ be the angle between $\mathbf{p}$ and $\mathbf{n}$. Then

$$|\mathbf{p}||\mathbf{n}|\cos\phi=\mathbf{p}\cdot\mathbf{n}=x(x+ay)+y(ax+y)=x^2+y^2+2axy=1,$$
$$|\mathbf{p}||\mathbf{n}|\sin\phi=|x(ax+y)-y(x+ay)|=|ax^2-ay^2|=a|y^2-x^2|,$$

where the second line is the size of the two-dimensional cross product (the area of the parallelogram spanned by $\mathbf{p}$ and $\mathbf{n}$), and $a > 0$.

Because the dot product is $+1 > 0$, ϕ is already acute (or zero), so $\phi = \theta$. Dividing the two lines,

$$\tan\theta = a\,|y^2 - x^2|.$$



The sketch (with $a = \frac{1}{2}$) shows the curve: since $0 < a < 1$ it is an ellipse with its axes along $y = \pm x$. The point P drawn is one where θ is largest.

Part (i)

The function $t \mapsto \arctan(t)$ is strictly increasing, so θ is a smooth increasing function of $f = y^2 - x^2$ wherever $f \neq 0$.

Ask yourself What about points where $y^2 = x^2$, where the modulus has a corner?

There $\theta = 0$. If θ has zero derivative at such a point then $a|f|$ divided by the change in x tends to 0, which forces $\mathrm{d}f/\mathrm{d}x = 0$ as well. So in every case

$$\frac{\mathrm{d}\theta}{\mathrm{d}x} = 0 \implies \frac{\mathrm{d}}{\mathrm{d}x}(y^2 - x^2) = 0 \implies y\frac{\mathrm{d}y}{\mathrm{d}x} = x.$$

Here we are using the gradient formula, so $ax + y \neq 0$ (we'll confirm below that this holds automatically). Substituting and multiplying by $ax + y$:

$$-y(x + ay) = x(ax + y) \implies -xy - ay^2 = ax^2 + xy \implies a(x^2 + y^2) + 2xy = 0.$$

That's (i).

Incidentally, if $ax + y = 0$ then (i) would give $ax^2(1 + a^2) - 2ax^2 = -a(1 - a^2)x^2 = 0$, so $x = y = 0$, which is not on the curve. So such points never cause trouble.

Part (ii)

Write $S = x^2 + y^2$. By (i), $2xy = -aS$. Putting this into the curve's equation,

$$S + a(-aS) = 1 \quad \Longrightarrow \quad S = \frac{1}{1-a^2}.$$

Hence

$$x^2 + y^2 + 2xy = S - aS = \frac{1-a}{1-a^2} = \frac{1}{1+a},$$

that is, $(1+a)(x^2+y^2+2xy) = 1$, as required.

Part (iii)

We need $|y^2 - x^2|$. Using $(y^2-x^2)^2 = (x^2+y^2)^2 - 4x^2y^2 = S^2 - (2xy)^2$,

$$(y^2-x^2)^2 = S^2 - a^2S^2 = (1-a^2)\cdot\frac{1}{(1-a^2)^2} = \frac{1}{1-a^2}.$$

Taking the positive square root (valid as $0 < a < 1$), $|y^2 - x^2| = 1/\sqrt{1-a^2}$, and so

$$\tan\theta = \frac{a}{\sqrt{1-a^2}}.$$

Checks

- Rotate axes: with $u = (x+y)/\sqrt{2}$ and $v = (x-y)/\sqrt{2}$ the curve becomes $(1+a)u^2 + (1-a)v^2 = 1$ and $y^2 - x^2 = -2uv$. By AM–GM, $1 \geq 2\sqrt{(1-a^2)}\,|uv|$, so $\tan\theta = 2a|uv| \leq a/\sqrt{1-a^2}$, with equality when $(1+a)u^2 = \frac{1}{2}$. That equality condition is exactly (ii), since $x^2+y^2+2xy = 2u^2$. So the stationary value is the maximum of θ.
- With $a = \frac{1}{2}$: $P \approx (1.1154,\ -0.2989)$ satisfies the curve equation and (i), and $\tan\theta = \frac{1}{2}|y^2 - x^2| \approx 0.5774 = 1/\sqrt{3}$, so $\theta = 30°$. A numerical scan of $\tan\theta$ round the ellipse for several values of a finds its maxima at exactly these values.
- As $a \to 0$ the curve becomes the unit circle, whose normal passes through O, so $\theta \to 0$, consistent with $\tan\theta \to 0$.

Common slips

- Using $\tan(\alpha - \beta)$ with the gradients of OP and the normal: it works, but it needs care with $x = 0$ and with signs, and the vector route avoids both.
- Forgetting why the angle found is the *acute* one: the dot product being $+1$ is what settles it.
- Squaring to find $|y^2 - x^2|$ and then forgetting to take the positive root.

Beyond STEP

Writing $x^2 + y^2 + 2axy$ as a quadratic form $\mathbf{x}^\mathsf{T} A\mathbf{x}$ with $A = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$, the axes of the ellipse are the eigenvectors of A, $(1,1)$ and $(1,-1)$, and the eigenvalues $1 \pm a$ give the semi-axes. This diagonalisation of symmetric matrices is a central idea of first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

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